Reviewer report: pandas 2.3.3 to 3.0.0rc0 (contains PR #56633)

Run `golden-pandas300rc0-upsetplot090`, tested against UpSetPlot 0.9.0. Report generated 2026-10-10 05:53 IST by reports/generate.py 1.0 (from Jonathan's render_report.py 0.1).

NEEDS ATTENTION

Recommended action: **Reject** Confidence: **High**

All required checks and evals passed, 3 runs per side gave identical results (stability old 1.0, new 1.0), 3 of 3 golden cases hold, tests ran without network.

Scope: this run compares two released versions, not a single PR's commits. Any change between the two versions can cause what is shown here.

Summary in plain English

- **What it says it does:** PR #56633 "CoW: Enable CoW by default and remove warning build" makes Copy-on-Write the default (PR title, github.com/pandas-dev/pandas/pull/56633, verified via the GitHub API on 2026-10-10).
- **What we observed:** 67 test(s) of UpSetPlot that passed with pandas 2.3.3 fail with pandas 3.0.0rc0, in 5 distinct error type(s). Same result in all 3 runs on each side. 6 test(s) already failed before the change and are not counted.
- **What you should do:** Do not approve as is. The change breaks UpSetPlot; see 'What changed in behavior' for what fails.

Claims vs reality

#	Claim (source)	Status	What we observed	Log line
C1	Copy-on-Write is enabled by default (PR #56633 title)	not testable	the dependent's tests do not check the pandas setting itself; they show its effect on dependent code	pytest-new-1.log:2104 «Invalid RGBA argument: nan»
-	Change not in the PR description (this run)	unmentioned change	36 test(s) now fail with: ValueError: Invalid RGBA argument: nan (linked to C1 by UpSetPlot issue #303 (github.com/jnothman/UpSetPlot/issues/303))	pytest-new-1.log:2104 «ValueError: Invalid RGBA argument: nan»
-	Change not in the PR description (this run)	unmentioned change	16 test(s) now fail with: AssertionError: DataFrame.index level [N] are different (no claim mentions this)	pytest-new-1.log:35 «AssertionError: DataFrame.index level [»
-	Change not in the PR description (this run)	unmentioned change	13 test(s) now fail with: ValueError: Do not know how to convert N NaN (no claim mentions this)	pytest-new-1.log:2510 «ValueError: Do not know how to convert»

#	Claim (source)	Status	What we observed	Log line
-	Change not in the PR description (this run)	unmentioned change	1 test(s) now fail with: ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not N #0000002e (no claim mentions this)	pytest-new-1.log:9372 «ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not»
-	Change not in the PR description (this run)	unmentioned change	1 test(s) now fail with: ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not N NaN (no claim mentions this)	pytest-new-1.log:9372 «ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not»

Citation check (PASS): 6 claim(s), every one backed by a real log line

What changed in behavior

67 real break(s) in 5 group(s), largest first.

#	Error after the change	Tests	Log line
1	ValueError: Invalid RGBA argument: nan	36	pytest-new-1.log:2104 «ValueError: Invalid RGBA argument: nan»
2	AssertionError: DataFrame.index level [N] are different	16	pytest-new-1.log:35 «AssertionError: DataFrame.index level [»
3	ValueError: Do not know how to convert N NaN	13	pytest-new-1.log:2510 «ValueError: Do not know how to convert»
4	ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not N #0000002e	1	pytest-new-1.log:9372 «ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not»
5	ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not N NaN	1	pytest-new-1.log:9372 «ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not»

Group 1: 36 test(s)

Error: ValueError: Invalid RGBA argument: nan

Affected tests: `test_add_stacked_bars[False-horizontal]`, `test_add_stacked_bars[False-vertical]`, `test_add_stacked_bars[True-horizontal]`, `test_add_stacked_bars[True-vertical]`, `test_add_stacked_bars_colors[<lambda>-expected4]`, `test_add_stacked_bars_colors[Pastel1-expected2]`, and 30 more (full list: class-real_break.txt)

Example: `upsetplot.tests.test_upsetplot::test_add_stacked_bars[False-horizontal]`. Before: passed, passed, passed (old runs). After: failed, failed, failed (new runs).

```
Before (old side): test passed, no error output

After (new side):
ValueError: Invalid RGBA argument: nan
```

Group 2: 16 test(s)

Error: AssertionError: DataFrame.index level [N] are different

Affected tests: `test_from_contents_vs_memberships[blah-iter-None]`, `test_from_contents_vs_memberships[blah-iter-data1]`, `test_from_contents_vs_memberships[blah-list-None]`, `test_from_contents_vs_memberships[blah-list-data1]`, `test_from_contents_vs_memberships[blah-set-None]`, `test_from_contents_vs_memberships[blah-set-data1]`, and 10 more (full list: class-real_break.txt)

Example: `upsetplot.tests.test_data::test_from_contents_vs_memberships[blah-iter-None]`. Before: passed, passed, passed (old runs). After: failed, failed, failed (new runs).

```
Before (old side): test passed, no error output

After (new side):
AssertionError: DataFrame.index level [0] are different
Attribute "dtype" are different
[left]:  object
[right]: bool
```

Group 3: 13 test(s)

Error: ValueError: Do not know how to convert N NaN

Affected tests: `test_two_sets[set20-set11]`, `test_two_sets[set20-set12]`, `test_two_sets[set21-set10]`, `test_two_sets[set21-set11]`, `test_two_sets[set21-set12]`, `test_two_sets[set21-set13]`, and 7 more (full list: class-real_break.txt)

Example: `upsetplot.tests.test_upsetplot::test_two_sets[set20-set11]`. Before: passed, passed, passed (old runs). After: failed, failed, failed (new runs).

```
Before (old side): test passed, no error output

After (new side):
ValueError: The truth value of a Series is ambiguous. Use a.empty, a.bool(), a.item(), a.any() or
a.all().
ValueError: Unrecognized linestyle: nan
ValueError: Do not know how to convert 0    NaN
1    NaN
2    NaN
3    NaN
Name: linestyles, dtype: object to dashes
```

Group 4: 1 test(s)

Error: ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not N #0000002e

Affected tests: `test_style_subsets_artists[horizontal]` (full list: class-real_break.txt)

Example: `upsetplot.tests.test_upsetplot::test_style_subsets_artists[horizontal]`. Before: passed, passed, passed (old runs). After: failed, failed, failed (new runs).

```
Before (old side): test passed, no error output

After (new side):
17         blue
18    #0000002e
19         blue
20         blue
21          NaN
22          NaN
23          NaN
Name: facecolors, dtype: object
```

Group 5: 1 test(s)

Error: ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not N NaN

Affected tests: `test_style_subsets_artists[vertical]` (full list: class-real_break.txt)

Example: `upsetplot.tests.test_upsetplot::test_style_subsets_artists[vertical]`. Before: passed, passed, passed (old runs). After: failed, failed, failed (new runs).

```
Before (old side): test passed, no error output

After (new side):
17    #0000002e
18          red
19    #0000002e
20    #0000002e
21    #0000002e
22    #0000002e
23    #0000002e
Name: facecolors, dtype: object
```

What went right

Status	Check	Detail
pass	Correct versions installed	old pandas==2.3.3, new pandas==3.0.0rc0, UpSetPlot==0.9.0 on both sides (freeze + import probe)
pass	Only the upstream version differs between sides	freeze diff: ['pandas==2.3.3', 'pandas==3.0.0rc0']
pass	Isolated environments	both venvs created by uv without system site-packages (checked sys.prefix and sys.path)
pass	Thread count pinned to 1	OMP_NUM_THREADS, OPENBLAS_NUM_THREADS, MKL_NUM_THREADS, NUMEXPR_NUM_THREADS, VECLIB_MAXIMUM_THREADS, BLIS_NUM_THREADS = 1 in both envs
pass	All runs completed	3 runs per side requested (minimum 3); JUnit XML present and no suite timeout on every run

Status	Check	Detail
pass	Old side healthy (baseline is trustworthy)	dependent imports on old: {'upsetplot': 'ok'}; old side passed [233, 233, 233] tests with collection errors [0, 0, 0]
pass	Repeatable results	share of tests with the same outcome in every run: old 1.0, new 1.0 (advisory threshold 0.99)
pass	No secrets in logs	scrubbed env; no credential-like strings found in logs
pass	Record format valid	record matches harness/schema/observation.schema.json
pass	Runs per side	3 (minimum 3), alternating old and new
pass	Guardrails respected	tests ran with no network; nothing was posted or filed by the runner
pass	Matches the prediction written before the run	break: 67 real breaks from Copy-on-Write (Invalid RGBA argument: nan) and 6 already broken (pytest.warns(None))
pass	Nathan's evals	6 of 7 eval checks passed (see 'Evals and guardrails')
pass	Every claim cites a log line	6 claim(s), every one backed by a real log line

What went wrong and how to fix it

Issues in the PR or release

Issue	How to fix
Group 1: 36 test(s) fail with: ValueError: Invalid RGBA argument: nan	Decide whether this change is intended. If yes, document it as a breaking change and tell UpSetPlot; if not, fix it in the PR. Reproduce with the rerun command below and the example test.
Group 2: 16 test(s) fail with: AssertionError: DataFrame.index level [N] are different	Decide whether this change is intended. If yes, document it as a breaking change and tell UpSetPlot; if not, fix it in the PR. Reproduce with the rerun command below and the example test.
Group 3: 13 test(s) fail with: ValueError: Do not know how to convert N NaN	Decide whether this change is intended. If yes, document it as a breaking change and tell UpSetPlot; if not, fix it in the PR. Reproduce with the rerun command below and the example test.
Group 4: 1 test(s) fail with: ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not N #0000002e	Decide whether this change is intended. If yes, document it as a breaking change and tell UpSetPlot; if not, fix it in the PR. Reproduce with the rerun command below and the example test.
Group 5: 1 test(s) fail with: ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not N NaN	Decide whether this change is intended. If yes, document it as a breaking change and tell UpSetPlot; if not, fix it in the PR. Reproduce with the rerun command below and the example test.

Issues in our test environment

Issue	How to fix
6 test(s) of UpSetPlot already fail before the change: TypeError: exceptions must be derived from Warning, not <class 'NoneType'> (6)	Not caused by this change; ignore for this decision. To clean up the baseline, pin the tool versions UpSetPlot expects or deselect these tests (record it in subset_note).

Trust checks

Check	Status	Detail
Required checks (gates)	7 of 7 passed	all passed
Advisory checks	2 of 2 passed	all passed
Evals (Nathan)	APPROVE	no failures or warnings
Golden cases (stored records)	3 of 3 hold	checked against the stored golden records (harness 1.0) with the eval gate's code, not re-run for this report. This run is itself one of the golden cases, so this check is partly circular for it
Golden cases (re-run)	not available	no golden re-run recorded in this record (golden_check)
Report citations	pass	6 claim(s), every one backed by a real log line
Flaky rate	0 of 251 (0.0%)	stability old 1.0, new 1.0
Network during tests	off	install: allowed (uv from PyPI/nightly index, git clone of the tag); tests: no network (unshare -rn, private network namespace)
Threads	pinned to 1	OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1, BLIS_NUM_THREADS=1
Machine	box	Intel(R) Xeon(R) Processor, 8 CPUs, 15.6 GB RAM, Linux x86_64, CI: not available
Report matches the record	consistent	break and import-failure counts recomputed from class-*.txt and JUnit equal the record
Prediction before the run	yes	break: 67 real breaks from Copy-on-Write (Invalid RGBA argument: nan) and 6 already broken (pytest.warns(None))

Golden case	Status	Detail
golden-sklearn160rc1-imblearn-import-failure	holds	golden-sklearn160rc1-imblearn-import-failure: all expected values hold
golden-pandas300rc0-upsetplot-cow	holds	golden-pandas300rc0-upsetplot-cow: all expected values hold
golden-clean-sklearn151-152-imblearn	holds	golden-clean-sklearn151-152-imblearn: all expected values hold

Evals and guardrails (Nathan)

Independent checks of this record by evals/checks.py and guardrails/schema.py against the shared schema `blast-radius/observation/v1` (written against rev 1 (sha256 5a3c3d36b773...); valid under current rev 2 (additive)). Eval recommendation: **APPROVE**. Any FAIL here makes the verdict COULD NOT VERIFY.

Check	Status	Detail	How to fix
schema	pass	record matches the shared schema blast-radius/observation/v1, mode v1 (schema reference verified by sha256)	-
versions_installed	pass	pandas 2.3.3 -> 3.0.0rc0; env differs only in pandas	-
runs_per_side	pass	3 alternating runs per side, identical counts within each side	-
single_thread	pass	OMP/OPENBLAS/MKL_NUM_THREADS=1	-
matches_expected	pass	outcome matches the prediction written before the run (break)	-
golden:golden-pandas300rc0-upsetplot-cow	pass	golden-pandas300rc0-upsetplot-cow: all expected values hold	-
llm_citations	skip	no Nemotron explanations in this record yet	-
report_citations	pass	6 claim(s), every one backed by a real log line	-

Reviewer checklist

☐ I read the verdict, the recommended action and the confidence.
☐ I checked the claims vs reality table for any 'false' or 'unmentioned change' rows, and opened at least one cited log line.
☐ I looked at the largest behavior-change group and its example error.
☐ I confirmed the required checks and Nathan's evals passed (Trust checks, Evals and guardrails).
☐ I am satisfied the environment issues (if any) do not change the decision.

Decision: ☐ Approve ☐ Approve with notes ☐ Reject ☐ Re-run Reviewer: _____ Date: _____

Rerun command

From the blast-radius repo root. The wrapper forces 3 alternating runs per side and the single-thread env, records machine_type, writes under ./data and never posts anything:

```
BR_RUNNER=/workspace/blast-radius-research/harness/br_run.py python -m harness.run \
  --upstream pandas \
  --old 2.3.3 \
  --new 3.0.0rc0 \
  --dependent UpSetPlot \
  --dep-version 0.9.0 \
  --tests repo:https://github.com/jnothman/UpSetPlot:upsetplot/tests \
  --extra-pkgs 'numpy==2.3.5 matplotlib==3.10.8 pytest==9.1.1' \
  --cpus 4-7 \
  --expected 'break: 67 real breaks from Copy-on-Write (Invalid RGBA argument: nan) and 6 already
broken (pytest.warns(None))' \
  --expected-class break \
  --id golden-pandas300rc0-upsetplot090-rerun
```

Needs Linux x86_64, uv 0.12 or newer, git and unshare.

Evidence links

What	Link (relative to this report)
Where the description comes from	PR title, github.com/pandas-dev/pandas/pull/56633, verified via the GitHub API on 2026-10-10
Claim C1	pytest-new-1.log:2104 «Invalid RGBA argument: nan»
Unmentioned change	pytest-new-1.log:2104 «ValueError: Invalid RGBA argument: nan»
Unmentioned change	pytest-new-1.log:35 «AssertionError: DataFrame.index level [»
Unmentioned change	pytest-new-1.log:2510 «ValueError: Do not know how to convert»
Unmentioned change	pytest-new-1.log:9372 «ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not»
Unmentioned change	pytest-new-1.log:9372 «ValueError: 'c' argument must be a color, a sequence of colors, or a sequence of numbers, not»
pytest summary, old side run 1	pytest-old-1.log:475 «6 failed, 233 passed, 12 skipped, 414 warnings in 6.59s»
pytest summary, new side run 1	pytest-new-1.log:10087 «73 failed, 166 passed, 12 skipped, 212 warnings in 10.62s»
Full record	record.json
Run arguments	args.json
Setup log	setup.log
Installed packages, old	freeze-old.txt
Installed packages, new	freeze-new.txt
Package difference	freeze-diff.txt
pytest logs	pytest-new-1.log, pytest-new-2.log, pytest-new-3.log, pytest-old-1.log, pytest-old-2.log, pytest-old-3.log
JUnit XML	junit-new-1.xml, junit-new-2.xml, junit-new-3.xml, junit-old-1.xml, junit-old-2.xml, junit-old-3.xml
Test lists per class	class-already_broken.txt, class-fixed_by_new.txt, class-flaky.txt, class-import_failure.txt, class-lost_tests.txt, class-real_break.txt

Appendix

Commits and versions

Item	Value
PR	not available
PR base / head / merge commit	not available

Item	Value
Upstream	pandas: old 2.3.3, new 3.0.0rc0 (requested 3.0.0rc0)
Upstream commits (old / new)	PyPI release wheels (no commit recorded)
Dependent	UpSetPlot 0.9.0
Tests from	repo tag 0.9.0 (https://github.com/jnothman/UpSetPlot, commit 7d889a9df7974509965a5ffd313e27764d43322a); selection: upsetplot/tests
Python	3.13.5
Environment difference	pandas==2.3.3, pandas==3.0.0rc0
Runner	harness 1.0, 4 workers, import mode importlib, per-test timeout 300 s, memory cap 6.0 GB
Record schema	v1; written against rev 1 (sha256 5a3c3d36b773...); valid under current rev 2 (additive)
Run by	Jonathan on box, 2026-10-10 05:36 IST
Hand check	not available
Upstream issue	not available
Per-test outcomes	not available
AI explanation	not yet wired

Timings

Total wall time 56.5 s; mean test time old 7.3 s, new 9.8 s.

Side	Run	Exit	Seconds	pytest summary
old	1	1	7.19	6 failed, 233 passed, 12 skipped, 414 warnings in 6.59s
new	1	1	11.4	73 failed, 166 passed, 12 skipped, 212 warnings in 10.62s
old	2	1	8.16	6 failed, 233 passed, 12 skipped, 414 warnings in 7.62s
new	2	1	8.75	73 failed, 166 passed, 12 skipped, 212 warnings in 8.20s
old	3	1	6.58	6 failed, 233 passed, 12 skipped, 414 warnings in 6.00s
new	3	1	9.16	73 failed, 166 passed, 12 skipped, 212 warnings in 8.55s

Glossary

Term	Meaning
Old side / new side	The same dependent library tested twice: once with the old version of the upstream library, once with the new one. Nothing else differs.
Upstream	The library being changed (for example pandas). The PR or release under review belongs to it.
Dependent	A library that uses the upstream. Its own tests are our test subject.

Term	Meaning
Real break	A test that passed in every old run and failed in every new run. This is a behavior change caused by the upstream change.
Import failure	A test module that cannot even be loaded on the new side, so none of its tests can run. Counted per module.
Lost tests	Tests that passed before but could not run after, because their module failed to load. Not counted as real breaks.
Already broken	Tests that failed before the change too. Not caused by the change.
Flaky	Tests whose result changes between runs on the same side. They are not trusted either way.
Error signature	The error message with numbers and file paths removed, so identical failures group together.
Gate	An automatic check by the runner that the run itself is trustworthy (right versions, same environment, enough runs). Required gates must pass for the result to count.
Eval	Nathan's independent check of the record (schema, versions, runs per side, threads, golden cases, citations). Any eval failure means the run cannot be trusted.
Golden case	A known example with a known answer. If the tool still gets the golden cases right, its answers on new cases are more believable.
Stability	Share of tests that gave the same result in every run on one side. 1.0 means fully repeatable.
Citation	A file name, line number and quoted text from a run log that backs a statement in the report.
JUnit XML	The machine-readable test result file pytest writes for each run.
Freeze	The exact list of installed packages and versions on each side.